

# JUBIN PHILIP

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Backend-focused software developer with experience building scalable backend systems, asynchronous processing pipelines, data-intensive applications, and AI-powered applications. Experienced in LLM integration, Retrieval-Augmented Generation (RAG), vector search, document processing, PostgreSQL/pgvector, and high-throughput backend systems. Experienced in leveraging Claude Code and Google Antigravity to accelerate development, debugging, and implementation.

## TECHNICAL EXPERTISE

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- AI/ML: RAG, LLM Integration, Embeddings, Semantic Search, PyTorch, Computer Vision, OCR, Model Training
- Backend: Python, FastAPI, Flask, Node.js, Express.js, NestJS, GraphQL, REST APIs
- Frontend: React.js, Next.js, HTML5, CSS3, JavaScript, TypeScript
- Databases & Data: PostgreSQL, pgvector, HNSW, MySQL, MongoDB, SQL, ETL, Web Scraping, PDF Parsing, Document Processing
- Development & Tools: Docker, Git, GitHub, Redis, RQ, Background Workers
- Languages: Python, JavaScript, TypeScript, PHP, Java, C

## PROFESSIONAL EXPERIENCE

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### Junior Software Developer

Mar 2025 – Present Digilaw Legal Software Private Limited, Kochi

- Designed and implemented Redis-based queue systems for asynchronous task execution, improving system scalability and responsiveness.
- Designed and implemented robust ETL pipelines for data extraction, transformation, and ingestion into PostgreSQL, ensuring reliable and structured data processing.
- Developed background worker services using Redis queues to handle heavy operations such as scraping, parsing, and bulk database updates.
- Decoupled synchronous APIs from long-running tasks using queue-based architecture, significantly reducing API latency and improving user experience.
- Implemented retry mechanisms, logging, and failure handling in worker pipelines to ensure fault tolerance and data consistency.
- Improved database performance by 30% through query optimization and indexing.
- Led migration from MySQL to PostgreSQL with zero data loss and improved query efficiency.
- Built secure REST APIs with JWT authentication and modular backend architecture.
- Developed and trained a PyTorch-based computer vision model for OCR text recognition using approximately 10,000 labeled images, including dataset preparation, train/validation/test splitting, model evaluation, and inference.
- Leveraged Claude Code and Google Antigravity to accelerate development, debugging, code generation, and implementation of backend and AI-related features.

### Software Developer Trainee

Aug 2024 – Feb 2025 Techfriar Technologies, Infopark, Kochi

- Developed full-stack applications using MERN and PERN stacks following industry best practices.
- Engineered efficient GraphQL APIs, reducing over-fetching and improving application performance.
- Built scalable backend systems using Node.js and PostgreSQL.
- Worked in agile teams to deliver production-ready applications with proper documentation.
- Optimized application performance and improved user experience for internal tools.

## PROJECTS

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### InkDex – AI Document Intelligence RAG Platform

FastAPI, PostgreSQL, pgvector Redis, RQ

- Under active development; deployed with FastAPI backend on Railway, Next.js frontend on Vercel, and Redis on Upstash.
- Built an AI-powered document question-answering platform using Retrieval-Augmented Generation (RAG) to answer user queries from uploaded documents.
- Implemented document processing pipelines including PDF extraction, recursive text chunking, metadata handling, and vector embedding generation using Sentence Transformers.
- Integrated Gemini LLM to generate context-aware answers using semantically retrieved document chunks.
- Implemented semantic vector search using PostgreSQL and pgvector with cosine similarity, Top-K retrieval, and similarity thresholding.

- Optimized vector retrieval using HNSW indexing with cosine distance for efficient approximate nearest-neighbor search.
- Designed the backend using FastAPI with modular Router, Service, and Repository layers, SQLAlchemy, and Alembic migrations.
- Implemented Redis and RQ background workers for asynchronous document processing, chunking, and embedding generation.
- Implemented JWT authentication, authorization, and document ownership controls to securely isolate user data.

#### Event Management Platform

Next.js, NestJS, GraphQL, PostgreSQL

- Developed a full-stack event management web application enabling users to create, manage, and book events seamlessly.
- Built a scalable backend using NestJS and GraphQL, supporting efficient data querying and real-time interaction patterns.
- Designed PostgreSQL schemas for handling events, users, bookings, and role-based access control.
- Implemented authentication and authorization using Firebase Auth with secure API handling.
- Developed responsive frontend using Next.js with optimized rendering and state management.
- Applied modular architecture and clean code principles to ensure maintainability and scalability.

#### Car Rental Platform

Next.js, Node.js, Express, GraphQL, PostgreSQL

- Developed a scalable car rental system with advanced filtering, booking management, and admin dashboard.
- Implemented OTP authentication using Twilio and email verification via Nodemailer.
- Built GraphQL APIs using Apollo Server and Prisma ORM.
- Integrated MinIO object storage, improving media load performance by 35
- Designed backend logic for handling concurrent bookings and transaction safety.
- Integrated Razorpay demo payment gateway for seamless and secure booking transactions.

## EDUCATION

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Master of Computer Applications  
CGPA: 8.31

2022 – 2024 Nirmala College, Muvattupuzha

Bachelor of Computer Applications  
CGPA: 7.4

2019 – 2022 Don Bosco Arts and Science College, Kannur

## CERTIFICATIONS

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- The Complete JavaScript Course – Udemy
- Python for Data Science – NPTEL
- MERN Stack Internship – Camerin Folks